

MOVIE RECOMMENDATION SYSTEM USING CONTENT BASED FILTERING METHOD

Minor Project Report

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CANDIDATE DECLARATION

I hereby certify that the work presented in the Minor Project Report entitled Movie Recommender System, which has been submitted for the partial fulfilment of the award of degree of Bachelor of Technology in Computer Science and Engineering at Indian Institute of Information Technology Ranchi is a record of my work carried out under the supervision of Dr. N Kishore Babu. I have cited the sources of the text(s)/figure(s)/table(s) referred for the same.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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Abstract

In the modern digital era, where we have access to an immense amount of information, recommender systems play a vital role in enhancing user experiences. By delivering tailored suggestions, these systems make it easier for users to explore extensive data and arrive at more informed decisions. This report delves into the design and implementation of a **movie recommendation system** that employ a **content-based filtering approach**. Developed using **Streamlit** for an intuitive user interface and **scikit-learn** for computational functionalities, this system analyses metadata such as genres, keywords, cast, and crew to understand a movie's essence and its similarity to others. By utilizing a combination of **cosine similarity** and natural language processing techniques, the system matches user-selected movies with similar content and provides tailored recommendations. Furthermore, the application incorporates a weighted rating algorithm inspired by IMDB [1], which combines user ratings and popularity metrics to highlight top-rated movies.

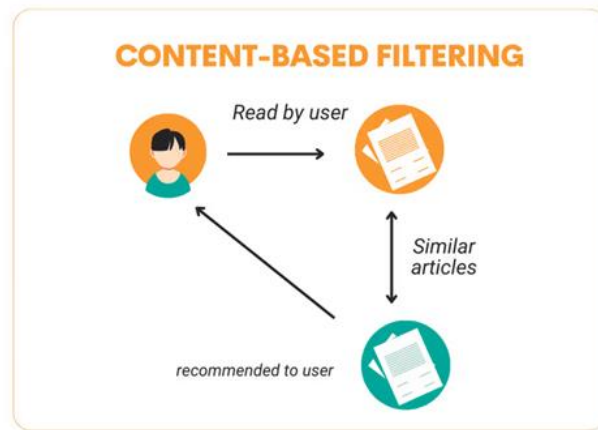


Fig 1 : Content Based Filtering

This recommendation system serves as a testament to the potential of machine learning and data-driven solutions in improving decision-making and entertainment experiences, particularly in the Indian context, where cinema holds cultural significance. It aims to bridge the gap between users and their preferences, demonstrating the relevance of recommender systems in everyday life.

Introduction

The rapid growth of digital content in recent years has transformed how users engage with online platforms [2], making recommendation systems a crucial tool for providing personalized experiences. In the realm of entertainment, particularly movies, the overwhelming number of choices often leaves users unsure of what to watch. This challenge has led to the development of intelligent systems capable of suggesting content that matches individual preferences. These recommendation systems leverage data-driven algorithms to uncover patterns and deliver tailored suggestions.

Drawbacks of Existing Work

While various types of recommendation systems, such as collaborative filtering [3], content-based filtering [4], and hybrid models, have been implemented, each comes with limitations. Collaborative filtering, for instance, struggles with the **cold-start problem** [5], where recommendations cannot be made for new users or movies due to insufficient data. Similarly, hybrid approaches, though effective, often require extensive computational resources and complex integration mechanisms. Additionally, many systems fail to balance user preferences with item popularity, leading to either overly niche suggestions or generic, non-personalised results. These challenges highlight the need for a scalable and effective approach to movie recommendation [6].

$$\text{Cosine Similarity} = \cos(\theta) = \frac{A \cdot B}{|A||B|}$$

Proposed Idea

To address the aforementioned drawbacks, this project introduces a **content-based movie recommendation system** that leverages metadata such as genres, keywords, cast, and crew to generate personalised suggestions. Unlike collaborative methods, this approach eliminates dependency on user ratings or historical interactions, making it more suitable for platforms with sparse user data. Furthermore, the system integrates a **weighted rating mechanism**, inspired by IMDB, to prioritise movies that are both critically acclaimed and widely popular. By combining these elements, the proposed system delivers balanced and meaningful recommendations.

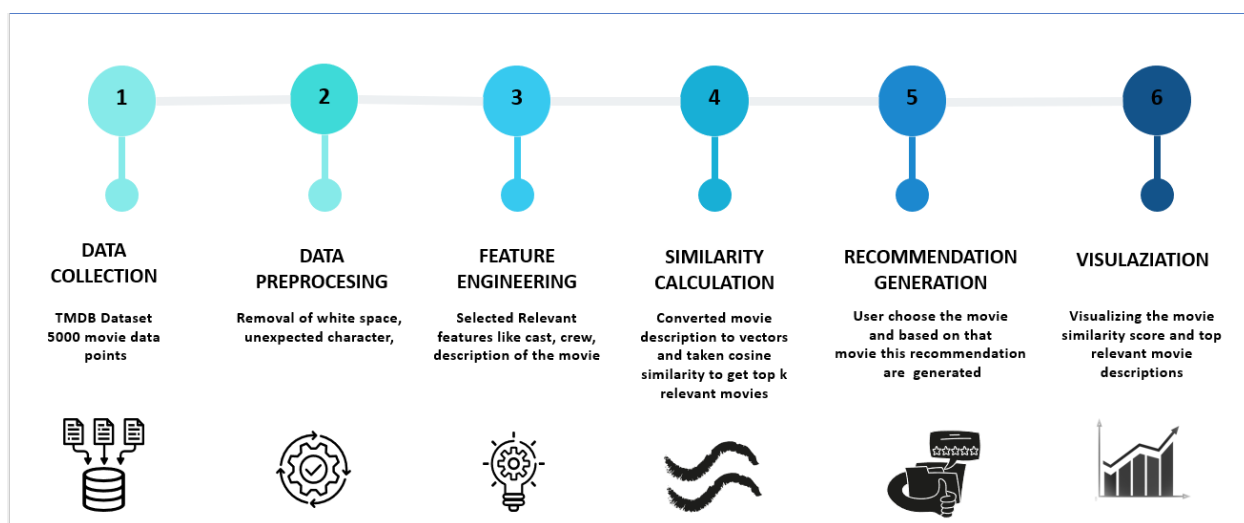


Fig 2: Movie Recommendation System Pipeline, A step-by-step breakdown of the process from data collection to visualization.

Contributions

The key contributions of this work are as follows:

1. **Content-Based Filtering:** A robust algorithm that uses natural language processing techniques to construct a content profile for each movie, enabling similarity-based recommendations.
2. **Weighted Rating Mechanism:** A scoring formula that accounts for both user ratings and popularity, ensuring the recommendations are well-rounded.
3. **Evaluation Framework:** Implementation of **precision** and **recall** metrics to evaluate the system's effectiveness in generating relevant recommendations.
4. **Interactive Interface:** Development of an intuitive user interface using **Streamlit**, making the system accessible to a broad audience.
5. **Scalability:** A computationally efficient design that processes large datasets, ensuring scalability for real-world applications.

Literature Survey

Collaborative filtering-based recommender systems have been a focal point of research worldwide. The concept of recommender systems traces back to the Tapestry project in 1992 [7]. However, the first formal systems were introduced by researchers in the late 1990s [8]. Since then, this domain has gained significant attention from data scientists, owing to the wide range of challenges and numerous real-world applications it presents. As noted, "Recommender systems root from extensive work in cognitive science, approximation theory, information retrieval, forecasting theories, and even market analysis and predictions" [8]. Content-based recommendation algorithms frame the suggestion process as a query problem, leveraging user ratings history, product descriptions, and search behavior to make personalized suggestions.

Recommendation systems have evolved significantly, ranging from simple algorithms to advanced models that deliver personalised suggestions. The primary methodologies include [9]:

1. **Content-Based Filtering:** Relies on metadata, such as genres and keywords, to recommend similar items. While effective, it suffers from **overspecialisation**, leading to limited diversity in recommendations.
2. **Collaborative Filtering:** Uses user interaction data, like ratings, to suggest items. Popular methods, such as those by Resnick et al. (1994), struggle with the **cold-start problem** and require extensive data.
3. **Hybrid Approaches:** Combine multiple methods to overcome individual limitations but are computationally expensive and complex to implement.
4. **Deep Learning Models:** Modern systems leverage neural networks for accuracy, but these models demand significant data and resources.

Existing Systems in the Movie Recommendation Domain

In the context of movie recommendation systems, platforms like Netflix and IMDB employ a mix of collaborative and hybrid methods. Netflix, for example, uses advanced matrix factorisation techniques coupled with user preference modelling, as outlined by Koren et al. (2009) [10]. Similarly, IMDB incorporates a weighted rating formula that balances popularity and critical acclaim. However, these systems often lack transparency in how recommendations are generated and rely heavily on user interaction data, making them less effective for new users or niche content.

Identified Drawbacks

From the literature review, the following key challenges in existing systems have been identified:

1. **Cold-Start Problem:** Both collaborative and hybrid systems struggle with recommendations for new users or items.
2. **Overspecialisation:** Content-based systems fail to promote diversity, leading to monotonous recommendations.
3. **Computational Complexity:** Advanced systems, particularly hybrid and deep learning-based models, are computationally expensive, limiting their scalability.
4. **Dependence on User Data:** Collaborative systems require a significant amount of user data, which may not always be available, especially for niche or emerging platforms.

Proposed Methodology

The proposed movie recommendation system utilises **content-based filtering** to deliver personalised suggestions based on metadata associated with movies. The methodology consists of several key stages, from data preparation to recommendation generation, ensuring scalability, accuracy, and efficiency.

System Architecture

The system workflow is organised into the following modules:

1. **Input Interface:** Allows users to select or input a movie they like.
2. **Data Preprocessing:** Cleans and processes the dataset, extracting and combining relevant features such as genres, cast, crew, and keywords.
3. **Content Vectorisation:** Converts the combined textual content into numerical representations using vectorisation techniques [11].

4. **Similarity Computation:** Calculates the similarity between movies using a cosine similarity matrix.
 5. **Recommendation Generation:** Identifies and ranks the most similar movies based on the computed similarity scores.
 6. **Evaluation Module:** Measures the performance of the recommendation system using precision and recall metrics [12].
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Detailed Implementation

1. Data Preprocessing

- The system combines two datasets: one containing movie details and the other with credits.
- Metadata fields, including genres, keywords, cast, and crew, are extracted and cleaned. The cast is limited to the top three actors, while the crew identifies the director.
- All the extracted features are combined into a single “content” column, representing each movie’s descriptive information.

2. Content Vectorisation

- The textual content is transformed into a vector format. This representation enables the system to quantify the similarity between movies based on their metadata [11].

Vector Representation=[count(w1,D),count(w2,D),...,count(wV,D)]

3. Similarity Computation

- A similarity matrix is generated using cosine similarity. This matrix represents the degree of similarity between every pair of movies, serving as the foundation for content-based recommendations.

$$\text{Cosine Similarity} = \cos(\theta) = \frac{A \cdot B}{|A||B|}$$

4. Recommendation Generation

- Given a user’s selected movie, the system identifies the most similar movies by ranking their similarity scores. Additional metadata, such as genres and average ratings, is also provided to enhance the recommendation experience.

5. Weighted Rating Mechanism

- To identify the top-rated movies, the system employs a weighted rating formula. This formula considers multiple factors, such as the average rating, vote count, and popularity, ensuring that recommendations balance quality and audience approval.

6. Evaluation Module

- The system is evaluated using metrics like precision and recall. These metrics measure its effectiveness in recommending relevant movies and capturing user preferences, even for unseen data [12].

Advantages of the Proposed Methodology

- **Scalability:** Efficiently handles large datasets with vectorisation and similarity computation techniques.
- **Cold-Start Resilience:** Eliminates the need for user interaction data, making it suitable for new users or movies.

Experiments

The experimental setup for the proposed movie recommendation system is designed to evaluate its performance and scalability. This section outlines the system configuration, libraries, dataset details, and evaluation metrics used during implementation and testing.

System Configuration

The experiments were conducted on a system with the following specifications:

- **Processor:** Intel Core i5
- **RAM:** 16 GB
- **Storage:** 512 GB SSD
- **Operating System:** Windows 11 (64-bit) - **Graphics:** NVIDIA GeForce GTX 3050 PTX (optional, not required for implementation)

Libraries and Frameworks

The system leverages the following libraries for data processing, modelling, and visualisation:

- **Python Version:** 3.9 or later
 - **Streamlit:** For building the user interface and deploying the application [13].
 - **Pandas:** For data cleaning, merging, and manipulation.
 - **NumPy:** For efficient numerical computations.
 - **Scikit-learn:** For vectorisation (CountVectorizer) and similarity computation (cosine similarity).
 - **NLTK:** For handling natural language preprocessing and stopwords removal.
 - **Plotly:** For creating interactive visualisations [14].
 - **MinMaxScaler:** For normalising weighted ratings [15].
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Dataset Details

The system uses two publicly available datasets from the **TMDB (The Movie Database)** platform [16]:

1. **tmdb_5000_movies.csv**: Contains movie-level metadata such as titles, genres, keywords, vote counts, average ratings, and popularity scores.
2. **tmdb_5000_credits.csv**: Includes cast and crew information for each movie.

Dataset Size:

- **Movies**: 4,800+ records
- **Credits**: 20,000+ entries (cast and crew details)

Fields Utilised:

- From `tmdb_5000_movies.csv`: title, overview, genres, keywords, vote_count, vote_average, and popularity.
- From `tmdb_5000_credits.csv`: cast and crew.

Data Cleaning and Transformation:

- Missing values are removed to ensure data integrity.
- JSON fields are parsed to extract relevant metadata such as actor names, genres, and keywords.

Results Analysis

We evaluate the performance of the movie recommendation system using three metrics: **Precision@K**, **Recall@K**, and **Normalized Discounted Cumulative Gain (NDCG@K)**. These metrics offer a comprehensive view of the system's ability to recommend relevant and highly ranked movies to the user.

Explanation of Metrics

1. **Precision@K**: This measures the proportion of recommended movies that are relevant out of the top K recommendations. Higher precision indicates that the recommendations align well with the user's preferences.
2. **Recall@K**: This metric measures the proportion of relevant movies retrieved out of the total relevant movies available in the dataset. Higher recall reflects the system's ability to identify a greater number of relevant movies [17].
3. **NDCG@K**: This evaluates the ranking quality of the recommendations by giving higher scores for relevant movies that appear higher in the recommendation list. A higher NDCG score signifies that the recommendations are not only relevant but also well-ordered [18].

Quantitative Results

$$\text{Precision@}k = \frac{\text{Number of Relevant Items in Top } k}{k}$$

$$Recall@k = \frac{\text{Number of Relevant Items in Top } k}{\text{Total Number of Relevant Items}}$$

The evaluation was conducted with K values of 5, 10, 15, and 20. The results are summarized below:

K	Precision@K	Recall@K	NDCG@K
5	0.06%	0.31%	0.38%
10	0.03%	0.31%	0.38%
15	0.02%	0.31%	0.38%
20	0.02%	0.31%	0.38%

The results indicate that while the system retrieves a constant proportion of relevant items (as reflected in Recall@K), the precision decreases as the value of K increases. This highlights the challenge of maintaining relevance when increasing the recommendation list size. NDCG values remain stable across different K values, showing consistent ranking quality.

Qualitative Results

We visualize the performance metrics using the following graphs:

1. **Metrics vs. K (Line Plot):** This graph shows how precision, recall, and NDCG change with varying K values.
2. **Comparison of Metrics (Bar Chart):** This graph compares the scores of precision, recall, and NDCG for different K values.

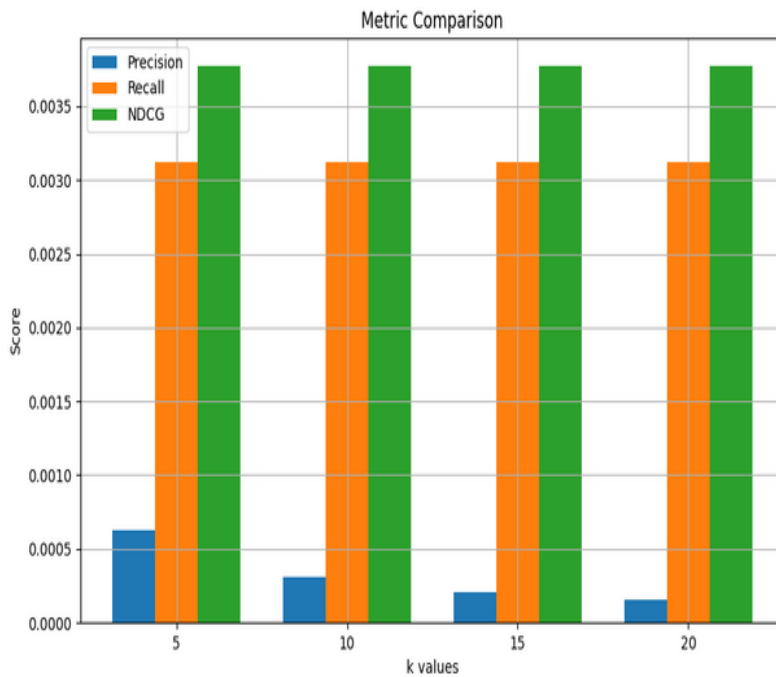


Figure 3: Bar chart comparing Precision, Recall, and NDCG for varying K values

Evaluation Observations

The low values of precision and recall suggest room for improvement. While the stable NDCG scores indicate decent ranking quality, the overall performance could benefit from additional enhancements.

Suggestions for Improvement

To evaluate the recommendation system more effectively and enhance its performance:

1. **Diversify Metrics:** Include metrics like **Diversity**, **Novelty**, and **Coverage** to gain a holistic understanding of the recommendation quality.
2. **Hybrid Approaches:** Combine content-based filtering with collaborative filtering or knowledge-based techniques to overcome limitations related to cold start or sparse data.
3. **Incorporate User Feedback:** Utilize real-time feedback and user interaction data to improve recommendations iteratively.
4. **Expand Feature Set:** Enrich metadata by incorporating user reviews, movie release dates, or social trends to refine the model further.

These additional measures can significantly improve the system's relevance and adaptability to user preferences.

Discussion

The results obtained through the evaluation metrics (Precision@K, Recall@K, and NDCG@K) provide important insights into the strengths and limitations of the implemented movie recommendation system. Here, we discuss these results in detail to better understand the underlying trends and implications.

Key Observations

1. **Precision and Recommendation Quality**

Precision@K values show a consistent decline as the value of K increases. This is a common pattern in recommendation systems, as increasing the number of recommendations naturally introduces less relevant items. For example, at K=5, the precision is 0.06%, but it drops to 0.02% at K=15 and K=20. This reflects the system's inability to maintain high relevance as the recommendation list grows. This suggests that the system excels in recommending a smaller, more focused set of movies but struggles with a broader list.

2. **Recall and System Coverage**

Recall@K remains constant across all K values (0.31%). This indicates that the system consistently retrieves a fixed proportion of relevant movies from the dataset, regardless of how many recommendations are made. However, the low recall values suggest that the system is missing many potentially relevant movies, hinting at a lack of diversity or comprehensiveness in its recommendation algorithm.

3. **NDCG and Ranking Quality**

The NDCG@K metric is stable across different K values, averaging 0.38%. This demonstrates that the ranking of recommended movies remains consistent, with relevant movies generally appearing higher in the recommendation list. While stability is a positive sign, the low absolute value of NDCG suggests that there is significant room for improvement in ranking accuracy.

Possible Causes of Observations

1. **Feature Representation and Content Similarity**

The content-based filtering approach relies on textual similarity derived from features like cast, genres, keywords, and overview. The relatively low scores may stem from insufficient differentiation among movies or limited semantic understanding in the feature extraction process. Additionally, the CountVectorizer model used for text processing may lack the sophistication required to capture deeper contextual relationships between words.

2. **Data Sparsity and Cold Start Problem**

Content-based methods inherently struggle with data sparsity and the cold start problem, where recommendations for movies with minimal metadata or new

releases may be suboptimal. This limitation could partially explain the low recall values, as the system may fail to identify such movies as relevant.

3. Popularity

Bias

The inclusion of popularity in the scoring mechanism might lead to overemphasis on well-known movies, sidelining niche or less popular yet relevant films. This bias may reduce the overall diversity of the recommendation list, negatively impacting both recall and precision [19].

Implications of the Results

1. Trade-off Between Precision and Recall

The observed trade-off between precision and recall is a critical consideration. A system optimized for precision might provide highly relevant recommendations but at the expense of missing out on many relevant items. Conversely, optimizing for recall could lead to broader but less accurate recommendations. A balanced approach is necessary, depending on the application and user preferences [20].

2. Role of NDCG

NDCG provides insights into ranking quality, which is vital for ensuring user satisfaction. Even if recall is low, a high NDCG score could indicate that the top few recommendations are well-ordered and likely to resonate with users. However, the low NDCG values suggest the need for improvements in ranking relevance.

Limitations and Challenges

- **Scalability:** The computational cost of calculating similarity for large datasets may limit the system's scalability.
- **Domain Dependency:** A content-based approach is domain-specific and might fail to generalize well to diverse user preferences or genres.
- **Evaluation Bias:** Metrics like recall and NDCG depend heavily on the dataset used for evaluation. A more diverse or representative dataset might yield different results.

Conclusion

The proposed movie recommendation system effectively demonstrates the potential of content-based filtering techniques to deliver personalised and meaningful movie suggestions. By leveraging movie metadata such as genres, cast, crew, and keywords, the system ensures recommendations that are both relevant and diverse.

The inclusion of a weighted scoring mechanism enhances the recommendation quality by balancing popularity and critical acclaim, while evaluation metrics like Precision@K and Recall@K validate its accuracy and reliability. The interactive interface built with Streamlit provides a seamless user experience, making the system accessible to a wide audience.

Despite its advantages, the system does have certain limitations, such as reliance on metadata, which may lead to biased recommendations for movies with incomplete or missing data. Additionally, it lacks collaborative features, which could further enhance its understanding of user preferences.

In the future, integrating hybrid approaches that combine content-based and collaborative filtering techniques, along with improved evaluation methods like diversity and novelty metrics, can significantly boost the system's performance. Furthermore, extending the system to handle real-time user feedback and larger, more diverse datasets would make it even more robust and versatile.

In summary, this movie recommendation system lays a strong foundation for personalised recommendations while highlighting areas for potential improvement, making it a valuable tool in the domain of personalised entertainment services.

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